Complex Numbers: Exponential Form

Questions 1–25

Question 1. Write each of the following in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$:

a) $3\mathrm{e}^{\frac{\pi}{3}\mathrm{i}}$
b) $5\mathrm{e}^{\frac{\pi}{4}\mathrm{i}}$
c) $2\mathrm{e}^{-\frac{4\pi}{3}\mathrm{i}}$
d) $7\mathrm{e}^{\frac{18\pi}{5}\mathrm{i}}$
e) $-4\mathrm{e}^{\frac{3\pi}{4}\mathrm{i}}$
f) $-2\mathrm{e}^{\frac{5\pi}{4}\mathrm{i}}$

Question 2. Write each of the following in the form $a + b\mathrm{i}$ where a and b are exact real constants:

a) $2\mathrm{e}^{\frac{\pi}{4}\mathrm{i}}$
b) $3\mathrm{e}^{\pi\mathrm{i}}$
c) $6\mathrm{e}^{\frac{7\pi}{3}\mathrm{i}}$
d) $2\mathrm{e}^{-\pi\mathrm{i}}$
e) $-4\mathrm{e}^{\frac{5\pi}{2}\mathrm{i}}$
f) $-\sqrt{2}\mathrm{e}^{-\frac{4\pi}{3}\mathrm{i}}$

Question 3. Write each of the following in the form $re^{i\theta}$, where $-\pi < \theta \leq \pi$, where r is given as an exact value:

a) $4\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$
b) $5\left(\cos(-\frac{\pi}{4}) + i\sin(-\frac{\pi}{4})\right)$
c) $\sqrt{2}\left(\cos(-\frac{4\pi}{5}) + i\sin(-\frac{4\pi}{5})\right)$
d) $5\left(\cos(-\frac{3\pi}{2}) + i\sin(-\frac{3\pi}{2})\right)$
e) $4\left(\cos\frac{\pi}{6} - i\sin\frac{\pi}{6}\right)$
f) $\sqrt{3}\left(\cos\frac{6\pi}{7} - i\sin\frac{6\pi}{7}\right)$
g) $\left(\cos(-\frac{\pi}{5}) - i\sin(-\frac{\pi}{5})\right)$
h) $8\left(\cos(-\frac{5\pi}{4}) - i\sin(-\frac{5\pi}{4})\right)$

Question 4. Write each of the following in the form $re^{i\theta}$, where $-\pi < \theta \leq \pi$, where r is given as an exact value, and θ is given as an exact value or rounded to three significant figures if appropriate:

a) $z = 2 + 2i$
b) $z = 4 - 4\sqrt{3}i$
c) $z = -2 + \sqrt{2}i$
d) $z = -5 - 12i$
e) $z = -7$
f) $z = 6i$
g) $z = 8$
h) $z = 3 + i$
i) $z = 1 - \sqrt{5}i$

Question 5. Write down the complex conjugate, z^*, of each of the following, giving your answer in the form $re^{i\theta}$, where $-\pi < \theta \leq \pi$.

a) $z = 2e^{\frac{\pi}{5}i}$
b) $z = 8e^{\frac{3\pi}{2}i}$
c) $z = -3e^{\frac{\pi}{7}i}$
d) $z = -6e^{\frac{16\pi}{7}i}$

Question 6. Find each of the following in the form $a + bi$ where a and b are real numbers. Use exact values for a and b where possible or give answers to three significant figures otherwise.

a) $e^{\frac{\pi}{4}i} \times e^{\frac{\pi}{2}i}$
b) $2e^{\frac{\pi}{3}i} \times \sqrt{3}e^{\frac{\pi}{5}i}$
c) $e^{2i} \times -2e^{5i}$
d) $e^{\frac{\pi}{4}i} \times e^{-\frac{\pi}{2}i} \times e^{\frac{\pi}{3}i}$
e) $2e^{-\frac{\pi}{6}i} \times \sqrt{2}e^{\frac{\pi}{3}i} \times e^{\frac{5\pi}{4}i}$

Question 7. Find each of the following in the form $a + b\mathrm{i}$ where a and b are real numbers. Use exact values for a and b where possible or give answers to three significant figures otherwise.

a) $\frac{\mathrm{e}^{\frac{\pi}{4}\mathrm{i}}}{\mathrm{e}^{\frac{\pi}{2}\mathrm{i}}}$

b) $\frac{2\mathrm{e}^{\frac{\pi}{3}\mathrm{i}}}{6\mathrm{e}^{\frac{\pi}{4}\mathrm{i}}}$

c) $\frac{\mathrm{e}^{3\mathrm{i}}}{5\mathrm{e}^{7\mathrm{i}}}$

d) $\frac{2\mathrm{e}^{\frac{\pi}{3}\mathrm{i}}}{\mathrm{e}^{-\frac{\pi}{4}\mathrm{i}}} \times 2\mathrm{e}^{-\frac{\pi}{2}\mathrm{i}}$

e) $\frac{\sqrt{2}\mathrm{e}^{-\frac{\pi}{4}\mathrm{i}}}{3\mathrm{e}^{\frac{\pi}{5}\mathrm{i}}} \times \mathrm{e}^{\frac{4\pi}{7}\mathrm{i}}$

Question 8. Write $(\cos 3\theta + \mathrm{i}\sin 3\theta)(\cos 2\theta + \mathrm{i}\sin 2\theta)$ in exponential form.

Question 9. Write each of the following in the form $re^{i\theta}$, where $-\pi < \theta \leq \pi$.

a) $\left(\cos\frac{3\pi}{7} + i\sin\frac{3\pi}{7}\right)\left(\cos\frac{4\pi}{7} + i\sin\frac{4\pi}{7}\right)$

b) $\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)\left(\cos(-\frac{\pi}{3}) + i\sin(-\frac{\pi}{3})\right)$

c) $3\left(\cos(-\frac{\pi}{4}) + i\sin(-\frac{\pi}{4})\right)\left(\cos\frac{3\pi}{8} + i\sin\frac{3\pi}{8}\right)$

d) $4\left(\cos(-\frac{\pi}{3}) + i\sin(-\frac{\pi}{3})\right)\left(\cos\frac{\pi}{4} - i\sin\frac{\pi}{4}\right)$

Question 10. Write

$$\frac{\cos 6\theta + i\sin 6\theta}{\cos 4\theta + i\sin 4\theta}$$

in exponential form.

Question 11. Write each of the following in the form $e^{i\theta}$, where $-\pi < \theta \leq \pi$:

a) $\frac{\cos\frac{2\pi}{5}+i\sin\frac{2\pi}{5}}{\cos\frac{4\pi}{5}+i\sin\frac{4\pi}{5}}$

b) $\frac{\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}}{\cos(-\frac{2\pi}{3})+i\sin(-\frac{2\pi}{3})}$

c) $\frac{\cos\frac{\pi}{5}+i\sin\frac{\pi}{5}}{\cos\frac{\pi}{3}-i\sin\frac{\pi}{3}}$

Question 12. Two complex numbers are given by $z = 2 - 2i$ and $w = 1 + \sqrt{3}i$

a) Write z in the form $re^{i\theta}$ where $r > 0$ and $-\pi < \theta \leq \pi$

b) Write w in the form $re^{i\theta}$ where $r > 0$ and $-\pi < \theta \leq \pi$

c) Hence show that if $z^2 + w^2 = a + bi$ where a and b are real constants, then

$$\frac{b}{a} = 4 - \sqrt{3}$$

Question 13. Given that $z = 3\mathrm{e}^{\frac{\pi}{4}\mathrm{i}}$ and $w = \sqrt{2}\mathrm{e}^{\frac{\pi}{3}\mathrm{i}}$, find the value of each of the following:

a) $|zw|$
b) $\left|\frac{z}{w}\right|$
c) $\arg(zw)$
d) $\arg\left(\frac{z}{w}\right)$

Question 14. Given that $z = \sqrt{3}\mathrm{e}^{-\frac{2\pi}{3}\mathrm{i}}$ and $w = 3\mathrm{e}^{\frac{\pi}{5}\mathrm{i}}$, find the value of each of the following:

a) $|zw|$
b) $\left|\frac{z}{w}\right|$
c) $\arg(zw)$
d) $\arg\left(\frac{z}{w}\right)$

Question 15. Given that $z_1 = r_1 e^{\theta_1 i}$ and $z_2 = r_2 e^{\theta_2 i}$, show that:

a) $|z_1 z_2| = |z_1||z_2|$

b) $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$

c) $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$

d) $\arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2)$

Question 16. It is given that $z = -6 + 6\sqrt{3}i$, $|w| = 4$ and $\arg w = -\frac{5\pi}{12}$. Find each of the following in the form $re^{i\theta}$, where $-\pi < \theta \leq \pi$:

a) zw

b) $\frac{z}{w}$

Question 17. Points A and B on an argand diagram are given by $z_1 = 4\mathrm{e}^{-\frac{\pi}{6}\mathrm{i}}$ and $z_2 = 4\mathrm{e}^{\frac{\pi}{6}\mathrm{i}}$ respectively. Point C is given by a third complex number z_3 ACB forms an isosceles triangle where $|z_3| = 4$.

a) Sketch z_1 and z_2 on the same Argand diagram.

b) Find the possible angle(s) $\angle ACB$

Question 18. Use $\mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta$ to show that

a) $\cos\theta = \frac{1}{2}(\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta})$

b) $\sin\theta = \frac{1}{2\mathrm{i}}(\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta})$

Question 19. Use $e^{i\theta} = \cos\theta + i\sin\theta$ to show that

a) $\sin^2\theta + \cos^2\theta = 1$

b) $2\sin\theta\cos\theta = \sin 2\theta$

c) $\cos^2\theta - \sin^2\theta = \cos 2\theta$

Question 20. By writing $\cos\theta + i\sin\theta$ as $e^{i\theta}$, show that

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$$

Question 21. Given that $z = a + b\mathrm{i}$ for real numbers a and b, show that $\mathrm{e}^{z^*} = (\mathrm{e}^z)^*$

Question 22. Show that $\mathrm{e}^{2\mathrm{i}\theta} - 1 = 2\mathrm{i}\mathrm{e}^{\mathrm{i}\theta}\sin\theta$

Question 23. Express $\left(1-\frac{1}{4}\mathrm{e}^{\mathrm{i}\theta}\right)\left(1-\frac{1}{4}\mathrm{e}^{-\mathrm{i}\theta}\right)$ in terms $\cos\theta$

Question 24. Express $(1-a\mathrm{e}^{\mathrm{i}\theta})(1-a\mathrm{e}^{-\mathrm{i}\theta})$ in terms of a and $\cos\theta$ where a is a real constant.

Question 25. Given that $z = 2e^{\frac{\pi}{6}i}$ and $w = 2e^{\frac{\pi}{3}i}$ show that

$$z + w = (\sqrt{2} + \sqrt{6})e^{\frac{\pi}{4}i}$$